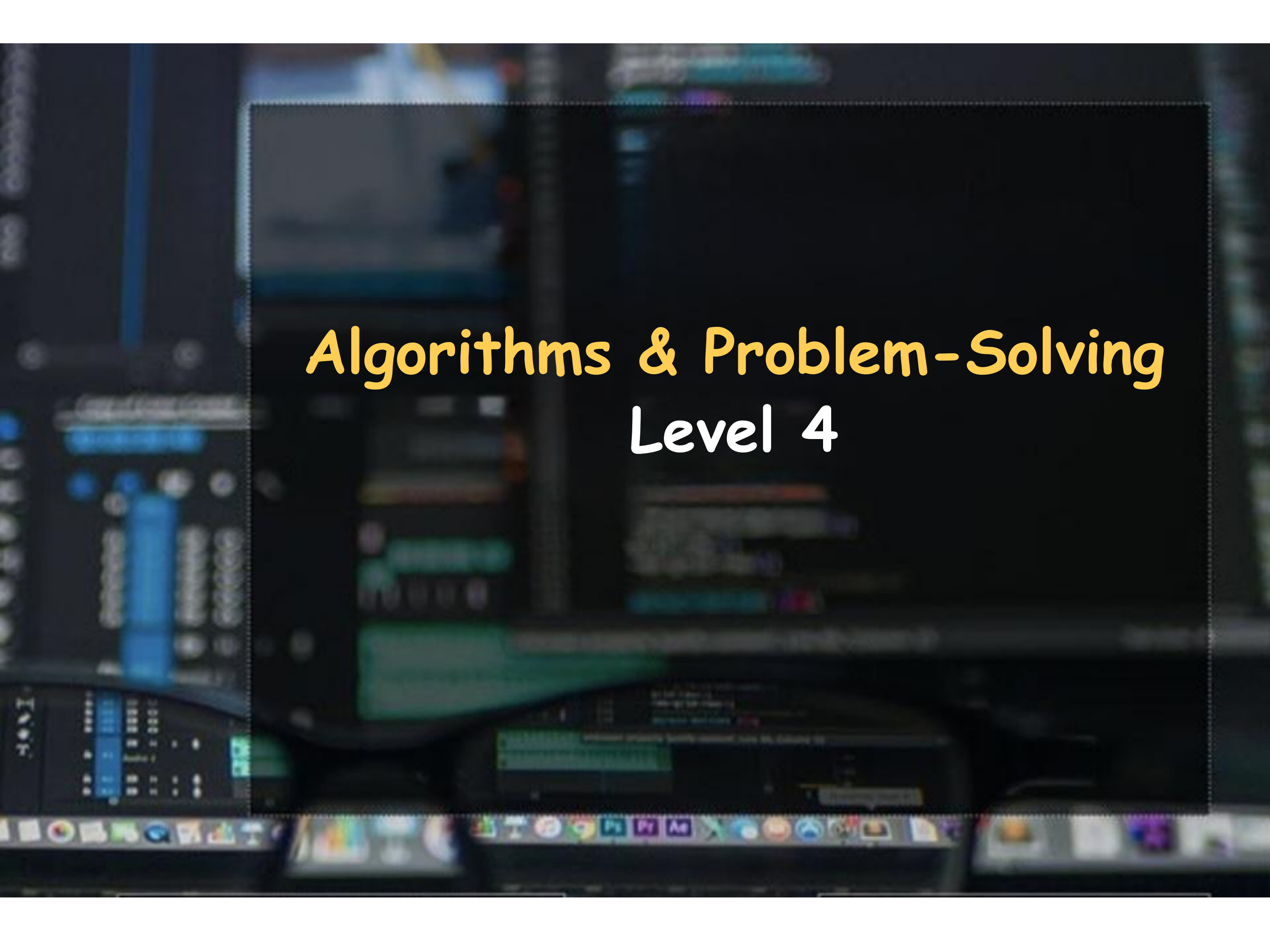
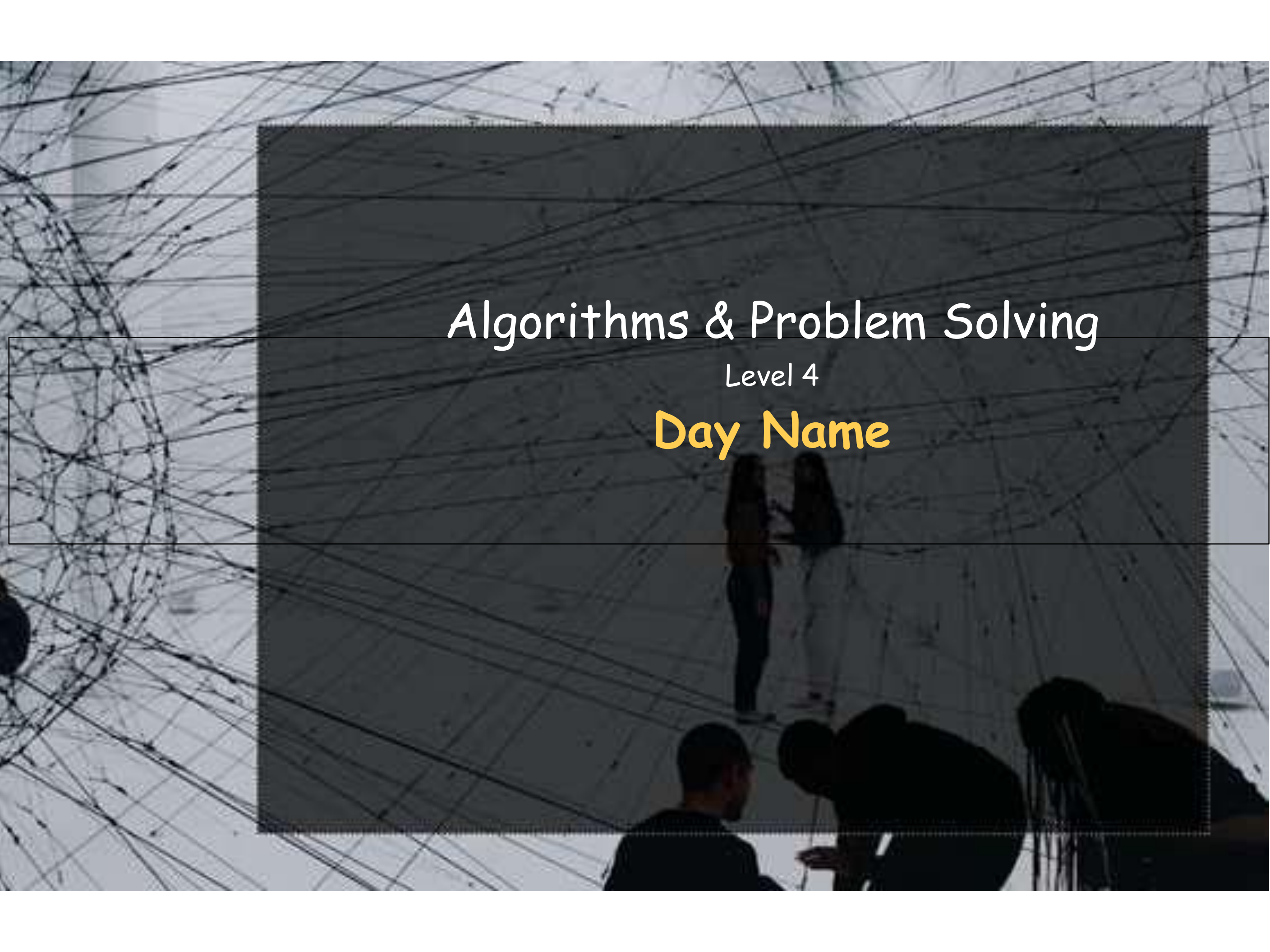




Algorithms & Problem-Solving Level 4

The background of the slide features a complex network diagram with numerous nodes and connecting lines, overlaid on a dark, moody image of people's silhouettes. A large, dark rectangular area is positioned in the center, serving as a backdrop for the text.

Algorithms & Problem Solving

Level 4

Day Name



Definition

- The goal is a program that reads the date (day, month, year) from the user and then displays:
 - The day's position in the week (from 0 to 6)
 - The day's name (e.g., Sun, Mon, Tue...)



Explanation

- The program relies on calculating the day's position within the week using a mathematical equation based on the *Gregorian* calendar, then converting this position into the day's name.

Procedures

1 The ReadYear() function

- Requires the user to enter the year and returns it.

2 The ReadMonth() function

- Reads the month number from the user.

3 The ReadDay() function

- Reads the day number from the user.

4 The IsLeapYear(int year) function

- Checks if the year is a leap year:

- Leap year if:

- $$(year \% 4 == 0 \ \&\& \ year \% 100 != 0) \ || \ (year \% 400 == 0)$$

5 NumberOfDaysInAMonth(int month, int year) function

- Returns the number of days in the month according to the year (29 in February if it's a leap year)

6 DayOfWeekOrder(int day, int month, int year) function

- Formula for calculating the day's position in the week:

- $$(day + y + (y / 4) - (y / 100) + (y / 400) + ((31 * m) / 12)) \% 7;$$

■ Where:

■ $a = (14 - \text{month}) / 12$

■ $y = \text{year} - a$

■ $m = \text{month} + (12 * a) - 2$

➔ The result is the day number:

0 = Sun, 1 = Mon, 2 = Tue, ...

⑦ The DayShortName(int dayOrder) function

■ Returns the day name using an array:

string arrDayNames[] = {"Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"};

⑧ The main() function

■ Reads the values, calculates the order, and then prints the date and day name



Example:

Please enter a year to check? 2022

Please enter a Month to check? 9

Please enter a Day? 20

Date: 20/9/2022

Day Order : 2

Day Name: Tue



Note:

- If the month or day is entered incorrectly, the result may be inaccurate because the program does not verify the date.

Summary

- ✓ The program uses:
 - Calculations to determine the day
 - Calendar equations (Gregorian formula)
 - Matrices to store day names

Thank you 😊